Meeting Analyzer

Grounded, Deterministic and Privacy-Conscious Meeting Intelligence

PPT Structure and Presentation Content

Presenter: _____ | Department: _____

1. Problem and Motivation

Why automated meeting analysis matters

Business problem: important decisions, commitments and deadlines are buried inside long audio/video recordings.

- Manual minute-taking is slow, incomplete and difficult to audit.
- Meeting conversations contain fillers, false starts, topic drift and implicit commitments.
- Cloud-only LLM solutions can hallucinate owners or deadlines and may expose sensitive data.
- Repeated analysis of the same transcript can produce different action-item counts.

Presentation message: the system is designed to be useful, explainable and repeatable - not just fluent.

2. Aim and Key Contributions

What the project delivers

Aim: convert recorded meetings into an evidence-linked, actionable and reviewable report.

- **End-to-end pipeline:** upload, audio extraction, transcription, NLP analysis and dashboard reporting.
- **Local-first processing:** FFmpeg, Faster-Whisper, spaCy, NLTK and scikit-learn reduce dependency on paid APIs.
- **Verbatim grounding:** actions, decisions and questions retain transcript evidence and source timestamps.
- **Deterministic reconciliation:** stable IDs, hashes and merge rules prevent duplicate rows and preserve edits.
- **Dual summarization:** extractive evidence-first minutes plus optional abstractive executive narrative.

3. End-to-End Workflow

The complete processing sequence

Input: recorded audio or video (MP4, MKV, MOV, WAV, MP3 or M4A)

Upload -> **Validate** -> **Extract Audio** -> **Transcribe** -> **NLP Preprocess**

NER | **Classification** | **Topics** -> **Actions / Decisions / Questions** -> **Summarization** -> **Dashboard + PDF**

Output: a canonical meeting report with summary, key takeaway, actions, decisions, questions, topics, entities and transcript evidence.

4. System Architecture

Modular backend and interactive frontend

Frontend: React + Vite, Tailwind CSS and Recharts provide upload, dashboard, history and analysis views.

Backend: Python + FastAPI exposes upload, analysis and report endpoints.

Processing services: audio, transcription, preprocessing, NER, classification, topics, actions, decisions and summarization.

Persistence: Supabase stores canonical intelligence; local recovery snapshots preserve inputs, candidates and previous reports.

Architecture principle: separate candidate generation from validation, reconciliation and persistence.

- AI proposes structured candidates.
- Local validation checks schema and evidence.
- Reconciliation merges new output with prior state and human edits.
- The dashboard displays the final canonical artifact.

5. NLP and Machine Learning Components

Why this is an NLP system

Speech-to-text: Faster-Whisper generates timestamped transcript segments from extracted audio.

Preprocessing: sentence segmentation, tokenization, normalization, stop-word handling and lemmatization.

Named Entity Recognition: spaCy identifies people, organizations, dates, locations, technologies and projects.

Dialogue-act classification: TF-IDF features with a scikit-learn classifier categorize sentences as action, decision, question, discussion or information.

Topic modeling: TF-IDF plus LDA/NMF highlights recurring themes and discussion areas.

Rule/dependency extraction: combines linguistic structure and validation rules to identify task-owner-deadline relationships.

6. Grounded Meeting Intelligence

From transcript sentences to actionable information

Actions: task, owner, deadline and status are extracted only when supported by transcript evidence.

Decisions: agreements and selected options are represented as concise decision statements.

Questions: raised questions are tracked as unresolved unless an answer is explicitly evidenced.

Evidence contract: supporting quotes must match the original transcript; source segment IDs provide navigation to timestamps.

Missing data policy: unknown owners, deadlines and priorities remain null instead of being guessed.

Trust benefit: every important report item can be inspected against the words that produced it.

7. Deterministic Re-analysis and Edit Protection

The core reliability feature

Problem: stochastic generation and changing context can make repeated analysis unstable.

- SHA-256 hashes represent canonical transcript input, model/configuration and schema versions.
- Stable IDs are derived from meeting, category, source segments and normalized payload.
- Exact or semantic matching merges equivalent insights and avoids duplicate rows.
- AI fields and user edits are stored separately; explicit human changes win during re-analysis.
- Incomplete or malformed provider responses fail safely and preserve the last successful report.

Result: re-analysis becomes a controlled state transition instead of replacing the report blindly.

8. Dual-Paradigm Summarization

Evidence-first minutes and executive narrative

Extractive mode (default for shareable evidence):

- TF-IDF sentence vectors and document centroid.
- Maximal Marginal Relevance reduces redundancy.
- Outcome weighting prioritizes actions and decisions.
- Selected sentences remain verbatim transcript excerpts.

Abstractive mode (optional):

- BART/T5/FLAN-T5 creates a fluent narrative from token-bounded chunks.
- Useful for executive readability, but requires human review.
- The system avoids silent truncation and reports incomplete generation.

9. User Interface and Reporting

How users interact with the system

Overview dashboard: duration, speaker information when available, action count, decision count and question count.

Transcript view: searchable transcript with evidence links and timestamp navigation.

Analytics view: topics, entities, classifications and meeting history trends.

Action and decision panels: owner-task-deadline rows, status, evidence and user-editable fields.

PDF export: ReportLab generates a shareable meeting report containing summary, transcript excerpts, actions, decisions and questions.

Demo flow: upload recording -> wait for analysis -> inspect evidence -> edit an action -> re-analyze -> verify edit is preserved.

10. Technology Stack

Implementation choices

Application: React, Vite, Tailwind CSS, Recharts

API: Python, FastAPI, Uvicorn

Audio and ASR: FFmpeg, Faster-Whisper, AV

NLP and ML: spaCy, NLTK, scikit-learn, NumPy, pandas

Transformers: Hugging Face Transformers, Torch, SentencePiece

Data and reporting: Supabase, ReportLab, JSON snapshots

Design goals: explainability, privacy, reproducibility, maintainability and low operating cost.

11. Evaluation and Results

Evidence from the consistency validation

Live consistency fixture: five fresh Groq analyses of the same meeting using the same input, schema, prompt and configuration.

Category	R1	R2	R3	R4	R5
Actions	2	2	2	2	2
Decisions	1	1	1	1	1
Questions	1	1	1	1	1
Follow-ups	0	0	0	0	0

- Minimum Jaccard overlap versus run 1: 100% for actions, decisions and questions.
- 47 backend semantic/reconciliation tests passed; 6 frontend data tests passed; production build passed.
- Unsupported entities in the cited evidence path were rejected rather than silently retained.

12. Limitations and Future Enhancements

What should be stated honestly

Current limitations:

- Current Whisper service does not infer speaker identity without diarization.
- English NLP models are used; multilingual quality requires further validation.
- The live consistency fixture is short and does not prove universal determinism.
- Lexical/evidence matching is not a full semantic-entailment verifier.
- Human review remains important for abstractive summaries and ambiguous commitments.

Future work:

- Speaker diarization and speaker-aware attribution.
- RAG and semantic meeting search.
- Ask-your-meeting question answering.
- Larger annotated evaluation datasets and confusion-matrix reporting.
- Calibration and broader multilingual testing.

13. Conclusion and Demonstration Plan

Close with the value proposition

Meeting Analyzer transforms unstructured recordings into trustworthy work artifacts.

- Local-first NLP provides transparent processing stages.
- Evidence grounding connects every insight back to the transcript.
- Deterministic reconciliation protects consistency and human edits.
- The dashboard and PDF report make results usable beyond the demo.

Recommended final slide prompt: ?How can grounded NLP make meeting outcomes more reliable??

Thank you